PRN – 260245920034

1. *List names of employees whose job is MANAGER.*
   mysql> select ename from emp where job = 'Manager';

```
mysql> select ename from emp where job = 'Manager';
+---------+
| ename   |
+---------+
| JONES   |
| BLAKE   |
| CLARK   |
| HOFFMAN |
| ALEX    |
+---------+
5 rows in set (0.00 sec)
```

2. *List names of employee having salary greater than equal to 3000.*
   mysql> select ename from emp where sal>=3000;

```
mysql> select ename from emp where sal>=3000;
+---------+
| ename   |
+---------+
| AARAV   |
| SCOTT   |
| KING    |
| FRED    |
| FORD    |
| HOFFMAN |
+---------+
6 rows in set (0.00 sec)
```

3. *List all employees having 'A' as second letter in their names.*
   mysql> select ename from emp where ename like '_A%';

```
mysql> select ename from emp where ename like '_A%';
+---------+
| ename   |
+---------+
| BANDISH |
| AARAV   |
| WARD    |
| MARTIN  |
| JAMES   |
| AARUSH  |
| NATASHA |
| SANGITA |
+---------+
8 rows in set (0.00 sec)
```

4. *List the Employee Number, Name, Hiredate and Job Title of the Employees in the Department 10.*
   mysql> select empno, ename, hiredate, job from emp where deptNo = 10;

```
mysql> select empno, ename, hiredate, job from emp where deptNo = 10;
+-------+--------+------------+------------+
| empno | ename  | hiredate   | job        |
+-------+--------+------------+------------+
|  7415 | AARAV  | 1981-12-31 | CLERK      |
|  7421 | THOMAS | 1981-07-19 | CLERK      |
|  7782 | CLARK  | 1981-06-09 | MANAGER    |
|  7839 | KING   | 1981-11-17 | PRESIDENT  |
|  7934 | MILLER | 1982-01-23 | CLERK      |
+-------+--------+------------+------------+
5 rows in set (0.00 sec)
```

5. *Display the details of those Employees who earn a monthly Salary of more than 2000.*
   mysql> select * from emp where sal>2000;

```
mysql>
mysql>
mysql>
mysql>
mysql> select * from emp where sal>2000;
+-------+---------+--------+---------------+------+------------+------+------+--------+---------+------------+------------+------------+----------+
| EMPNO | ENAME   | GENDER | JOB           | MGR  | HIREDATE   | SAL  | COMM | DEPTNO | BONUSID | USER NAME  | PWD        | PHONE      | isActive |
+-------+---------+--------+---------------+------+------------+------+------+--------+---------+------------+------------+------------+----------+
|  7415 | AARAV   | M      | CLERK         | 7902 | 1981-12-31 | 3350 | NULL |     10 |    NULL | AARAV      | NULL       | 7032300084 |        0 |
|  7566 | JONES   | M      | MANAGER       | 7839 | 1981-04-02 | 2975 | NULL |     20 |       4 | HONEYCOMB  | a12recmpm  | 7132300039 |        1 |
|  7698 | BLAKE   | M      | MANAGER       | 7839 | 1981-05-01 | 2850 | NULL |     30 |       1 | BIG BEN    | sales@2017 | 7132300027 |        1 |
|  7782 | CLARK   | M      | MANAGER       | 7839 | 1981-06-09 | 2450 | NULL |     10 |       3 | CLARK      | r50mpm     | 7032300001 |        1 |
|  7788 | SCOTT   | M      | ANALYST       | 7566 | 1982-12-09 | 3000 | NULL |     20 |       3 | WHITE SAND | a12recmpm  | NULL       |        1 |
|  7839 | KING    | M      | PRESIDENT     | NULL | 1981-11-17 | 5000 | NULL |     10 |       1 | KING       | r50mpm     | 7132300081 |        1 |
|  7840 | FRED    | M      | vice PRESIDENT | 7839 | 1983-08-09 | 5000 | NULL |     20 |       1 | SAND STONE | a12recmpm  | NULL       |        1 |
|  7902 | FORD    | M      | ANALYST       | 7566 | 1981-12-03 | 3000 | NULL |     20 |       4 | CONBRIO    | a12recmpm  | NULL       |        0 |
|  7919 | HOFFMAN | M      | MANAGER       | 7566 | 1982-03-24 | 4150 | NULL |     30 |       3 | INTERVAL   | sales@2017 | NULL       |        1 |
|  7920 | GRASS   | M      | SALESMAN      | 7919 | 1980-02-14 | 2575 | 2700 |     30 |       5 | GRASS      | sales@2017 | NULL       |        1 |
+-------+---------+--------+---------------+------+------------+------+------+--------+---------+------------+------------+------------+----------+
10 rows in set (0.00 sec)

mysql>
```

6. *Display the names of all Employees from Department 10 and 20 whose name start with "A" and ends with "B".*
   mysql> select ename from emp where deptno in (10, 20) and ename like 'A%B';
   Empty set (0.00 sec)

7. *Display the details of the employees who have joined on 21-DEC-1990.*
   mysql> select * from emp where hiredate = '1990-12-21';
   Empty set (0.00 sec)

8. *Display the Employee Code, Name, Job & Annual Salary for all Employees belonging to Department Number 10.*
   mysql> select empno as EmployeeCode, ename as Name, job as JobTitle, sal * 12 as AnnualSalary from emp where deptno = 10;

9. *Display the details of the Employees who earn an Annual salary of more than 25000. Sort the records in the descending order of Salary.*
   mysql> select * from emp where sal * 12 > 25000 order by sal desc;

```
mysql> select * from emp where sal * 12 > 25000 order by sal desc;
+-------+---------+--------+----------------+------+------------+------+------+--------+---------+------------+------------+------------+----------+
| EMPNO | ENAME   | GENDER | JOB            | MGR  | HIREDATE   | SAL  | COMM | DEPTNO | BONUSID | USER NAME  | PWD        | PHONE      | isActive |
+-------+---------+--------+----------------+------+------------+------+------+--------+---------+------------+------------+------------+----------+
|  7839 | KING    | M      | PRESIDENT      | NULL | 1981-11-17 | 5000 | NULL |     10 |       1 | KING       | r50mpm     | 7132300081 |        1 |
|  7840 | FRED    | M      | vice PRESIDENT | 7839 | 1983-08-09 | 5000 | NULL |     20 |       1 | SAND STONE | a12recmpm  | NULL       |        1 |
|  7919 | HOFFMAN | M      | MANAGER        | 7566 | 1982-03-24 | 4150 | NULL |     30 |       3 | INTERVAL   | sales@2017 | NULL       |        1 |
|  7415 | AARAV   | M      | CLERK          | 7902 | 1981-12-31 | 3350 | NULL |     10 |    NULL | AARAV      | NULL       | 7032300084 |        0 |
|  7788 | SCOTT   | M      | ANALYST        | 7566 | 1982-12-09 | 3000 | NULL |     20 |       3 | WHITE SAND | a12recmpm  | NULL       |        1 |
|  7902 | FORD    | M      | ANALYST        | 7566 | 1981-12-03 | 3000 | NULL |     20 |       4 | CONBRIO    | a12recmpm  | NULL       |        0 |
|  7566 | JONES   | M      | MANAGER        | 7839 | 1981-04-02 | 2975 | NULL |     20 |       4 | HONEYCOMB  | a12recmpm  | 7132300039 |        1 |
|  7698 | BLAKE   | M      | MANAGER        | 7839 | 1981-05-01 | 2850 | NULL |     30 |       1 | BIG BEN    | sales@2017 | 7132300027 |        1 |
|  7920 | GRASS   | M      | SALESMAN       | 7919 | 1980-02-14 | 2575 | 2700 |     30 |       5 | GRASS      | sales@2017 | NULL       |        1 |
|  7782 | CLARK   | M      | MANAGER        | 7839 | 1981-06-09 | 2450 | NULL |     10 |       3 | CLARK      | r50mpm     | 7032300001 |        1 |
+-------+---------+--------+----------------+------+------------+------+------+--------+---------+------------+------------+------------+----------+
10 rows in set (0.00 sec)

mysql>
```

10. *Display the names of Employees whose names contain the character "F".*
    mysql> select ename from emp where ename like '%F%';

```
mysql> select ename from emp where ename like '%F%';
+---------+
| ename   |
+---------+
| FRED    |
| FORD    |
| HOFFMAN |
+---------+
3 rows in set (0.00 sec)
```

11. *Display the Employee Code, Name, Job & Annual Salary for all Employees belonging to Department Number 10.*
    mysql> select empno as EmployeeCode, ename as Name, job as JobTitle, sal * 12 as AnnualSalary from emp where deptno = 10;

```
mysql> select empno as EmployeeCode,  ename as Name,  job as JobTitle,  sal * 12 as AnnualSalary from emp where deptno = 10;
+--------------+--------+-----------+--------------+
| EmployeeCode | Name   | JobTitle  | AnnualSalary |
+--------------+--------+-----------+--------------+
|         7415 | AARAV  | CLERK     |        40200 |
|         7421 | THOMAS | CLERK     |        21000 |
|         7782 | CLARK  | MANAGER   |        29400 |
|         7839 | KING   | PRESIDENT |        60000 |
|         7934 | MILLER | CLERK     |        15600 |
+--------------+--------+-----------+--------------+
5 rows in set (0.00 sec)
```

12. *Display the details of the Employees who earn an Annual salary of more than 25000. Sort the records in the descending order of Salary.*
    mysql> select * from emp where sal * 12 > 25000 order by sal desc;


```
mysql> select empno as EmployeeCode,  ename as Name,  job as JobTitle,  sal * 12 as AnnualSalary from emp where deptno = 10;
+--------------+--------+-----------+--------------+
| EmployeeCode | Name   | JobTitle  | AnnualSalary |
+--------------+--------+-----------+--------------+
|         7415 | AARAV  | CLERK     |        40200 |
|         7421 | THOMAS | CLERK     |        21000 |
|         7782 | CLARK  | MANAGER   |        29400 |
|         7839 | KING   | PRESIDENT |        60000 |
|         7934 | MILLER | CLERK     |        15600 |
+--------------+--------+-----------+--------------+
5 rows in set (0.00 sec)
```

13. *Display the names of Employees whose names contain the character "F".*

mysql> select ename from emp where ename like '%F%';

```
+---------+
| ename   |
+---------+
| FRED    |
| FORD    |
| HOFFMAN |
+---------+
3    rows in set (0.00 sec)
```

14. *Display the Names and Numbers of all Departments in the descending order of the Department Names.*
mysql> select dname, deptno from dept order by dname desc;

```
mysql> select dname, deptno from dept order by dname desc;
+------------+--------+
| dname      | deptno |
+------------+--------+
| SALES      |     30 |
| RESEARCH   |     20 |
| OPERATIONS |     40 |
| IT         |     60 |
| HR         |     50 |
| ACCOUNTING |     10 |
+------------+--------+
6 rows in set (0.00 sec)
```

15. *Display the Name, Employee Number, Designation and Salary of those Employees who earn a monthly salary of not less than 1000 and not more than 2000.*

mysql> select ename as Name, empno as EmployeeNumber, job as Designation, sal as Salary from emp where sal between 1000 and 2000;

```
mysql> select ename as Name, empno as EmployeeNumber,  job as Designation, sal as Salary from emp where sal between 1000 and 2000;
+----------+----------------+--------------------+--------+
| Name     | EmployeeNumber | Designation        | Salary |
+----------+----------------+--------------------+--------+
| GITA     |           6001 | Compliance officers |   1750 |
| VRUSHALI |           6129 | Compliance officers |   1750 |
| SHARMIN  |           6473 | Public Relation     |   1750 |
| BANDISH  |           6781 | Public Relation     |   1750 |
| THOMAS   |           7421 | CLERK               |   1750 |
| ALLEN    |           7499 | SALESMAN            |   1600 |
| WARD     |           7521 | SALESMAN            |   1250 |
| MARTIN   |           7654 | SALESMAN            |   1250 |
| TURNER   |           7844 | SALESMAN            |   1500 |
| ADAMS    |           7876 | CLERK               |   1100 |
| Ruhan    |           7888 | Network             |   2000 |
| Neel     |           7889 | vice PRESIDENT      |   1000 |
| MILLER   |           7934 | CLERK               |   1300 |
| AARUSH   |           7945 | SALESMAN            |   1350 |
| ALEX     |           7949 | MANAGER             |   1250 |
| NATASHA  |           8433 | CLERK               |   1750 |
| SANGITA  |           9400 | Compliance officers |   1750 |
| SUPRIYA  |           9473 | ANALYST             |   1750 |
+----------+----------------+--------------------+--------+
18 rows in set (0.00 sec)
```

16. Find out the Bonus (Bonus 1.1% of Annual Salary) that every person will receive at the end of the year.

```
select empno as EmployeeCode,
    ename as Name,
    job as Designation,
    sal * 12 as AnnualSalary,
    sal * 12 * 0.011 as Bonus
from emp;
```

```
+--------------+----------+---------------------+--------------+---------+
| EmployeeCode | Name     | Designation         | AnnualSalary | Bonus   |
+--------------+----------+---------------------+--------------+---------+
|         6001 | GITA     | Compliance officers |        21000 | 231.000 |
|         6129 | VRUSHALI | Compliance officers |        21000 | 231.000 |
|         6473 | SHARMIN  | Public Relation     |        21000 | 231.000 |
|         6781 | BANDISH  | Public Relation     |        21000 | 231.000 |
|         7369 | SMITH    | CLERK               |         9600 | 105.600 |
|         7415 | AARAV    | CLERK               |        40200 | 442.200 |
|         7421 | THOMAS   | CLERK               |        21000 | 231.000 |
|         7499 | ALLEN    | SALESMAN            |        19200 | 211.200 |
|         7521 | WARD     | SALESMAN            |        15000 | 165.000 |
|         7566 | JONES    | MANAGER             |        35700 | 392.700 |
|         7654 | MARTIN   | SALESMAN            |        15000 | 165.000 |
|         7698 | BLAKE    | MANAGER             |        34200 | 376.200 |
|         7782 | CLARK    | MANAGER             |        29400 | 323.400 |
|         7788 | SCOTT    | ANALYST             |        36000 | 396.000 |
|         7839 | KING     | PRESIDENT           |        60000 | 660.000 |
|         7840 | FRED     | vice PRESIDENT      |        60000 | 660.000 |
|         7844 | TURNER   | SALESMAN            |        18000 | 198.000 |
|         7876 | ADAMS    | CLERK               |        13200 | 145.200 |
|         7888 | Ruhan    | Network             |        24000 | 264.000 |
|         7889 | Neel     | vice PRESIDENT      |        12000 | 132.000 |
|         7900 | JAMES    | CLERK               |        11400 | 125.400 |
|         7902 | FORD     | ANALYST             |        36000 | 396.000 |
|         7919 | HOFFMAN  | MANAGER             |        49800 | 547.800 |
|         7920 | GRASS    | SALESMAN            |        30900 | 339.900 |
|         7934 | MILLER   | CLERK               |        15600 | 171.600 |
|         7945 | AARUSH   | SALESMAN            |        16200 | 178.200 |
|         7949 | ALEX     | MANAGER             |        15000 | 165.000 |
|         8433 | NATASHA  | CLERK               |        21000 | 231.000 |
|         9400 | SANGITA  | Compliance officers |        21000 | 231.000 |
|         9473 | SUPRIYA  | ANALYST             |        21000 | 231.000 |
+--------------+----------+---------------------+--------------+---------+
30 rows in set (0.00 sec)
```

17. Display the Name, Job and Dept No. of all the Employees who either work in Department 30 or in Department 40.

```
select ename as Name,
    job as JobTitle,
    deptno as DepartmentNumber
from emp
where deptno in (30, 40);
```

```
mysql> select ename as Name,
    ->        job as JobTitle,
    ->        deptno as DepartmentNumber
    -> from emp
    -> where deptno in (30, 40);
+---------+---------------------+------------------+
| Name    | JobTitle            | DepartmentNumber |
+---------+---------------------+------------------+
| GITA    | Compliance officers |               30 |
| ALLEN   | SALESMAN            |               30 |
| WARD    | SALESMAN            |               30 |
| MARTIN  | SALESMAN            |               30 |
| BLAKE   | MANAGER             |               30 |
| TURNER  | SALESMAN            |               30 |
| JAMES   | CLERK               |               30 |
| HOFFMAN | MANAGER             |               30 |
| GRASS   | SALESMAN            |               30 |
| AARUSH  | SALESMAN            |               30 |
| ALEX    | MANAGER             |               30 |
| NATASHA | CLERK               |               30 |
+---------+---------------------+------------------+
12 rows in set (0.00 sec)
```

18. Display the Employee Number, Name for Clerks and Managers who's Salary is more than 1500.

mysql> select empno as EmployeeNumber,

-> ename as Name

-> from emp

-> where job in ('CLERK', 'MANAGER')

-> and sal > 1500;

```
mysql> select empno as EmployeeNumber,
    ->        ename as Name
    -> from emp
    -> where job in ('CLERK', 'MANAGER')
    ->   and sal > 1500;
+----------------+---------+
| EmployeeNumber | Name    |
+----------------+---------+
|           7415 | AARAV   |
|           7421 | THOMAS  |
|           7566 | JONES   |
|           7698 | BLAKE   |
|           7782 | CLARK   |
|           7919 | HOFFMAN |
|           8433 | NATASHA |
+----------------+---------+
7 rows in set (0.00 sec)
```

19. Display the Name, Annual Salary and Commission as "Commission Earned" of all employees whose monthly salary is greater than their Commission. The output should be in decreasing order of salary. If two or more employees have same salary, then sort them by name within the highest order of salary.

select ename as Name,

*sal * 12 as AnnualSalary,*

comm as "Commission Earned"

from emp

where sal > comm

order by sal desc, ename asc;

```
mysql> select ename as Name,
    ->        sal * 12 as AnnualSalary,
    ->        comm as "Commission Earned"
    -> from emp
    -> where sal > comm
    -> order by sal desc, ename asc;
+--------+--------------+-------------------+
| Name   | AnnualSalary | Commission Earned |
+--------+--------------+-------------------+
| ALLEN  |        19200 |               300 |
| TURNER |        18000 |                 0 |
| ALEX   |        15000 |               500 |
| WARD   |        15000 |               500 |
+--------+--------------+-------------------+
4 rows in set (0.00 sec)
```

20. Display the Name and Job of sales executives or managers whose names end with the letter 'S'.

select ename as Name,

job as JobTitle

from emp

where job in ('SALESMAN', 'MANAGER')

and ename like '%S';

```
mysql> select ename as Name,
    ->        job as JobTitle
    -> from emp
    -> where job in ('SALESMAN', 'MANAGER')
    ->   and ename like '%S';
+-------+----------+
| Name  | JobTitle |
+-------+----------+
| JONES | MANAGER  |
| GRASS | SALESMAN |
+-------+----------+
2 rows in set (0.00 sec)
```

21. Display all the Names of the Employees whose Salary is either more than 1000 but not more than 2200.

mysql> select ename as Name from emp where sal > 1000 and sal <= 2200;

```
mysql> select ename as Name from emp where sal > 1000  and sal <= 2200;
+----------+
| Name     |
+----------+
| GITA     |
| VRUSHALI |
| SHARMIN  |
| BANDISH  |
| THOMAS   |
| ALLEN    |
| WARD     |
| MARTIN   |
| TURNER   |
| ADAMS    |
| Ruhan    |
| MILLER   |
| AARUSH   |
| ALEX     |
| NATASHA  |
| SANGITA  |
| SUPRIYA  |
+----------+
17 rows in set (0.00 sec)
```

22. *Select Name & Salary of employees who are clerks.*

mysql> select ename as Name, sal as Salary from emp where job = 'CLERK';

```
mysql>
mysql>
mysql> select ename as Name,  sal as Salary from emp where job = 'CLERK';
+---------+--------+
| Name    | Salary |
+---------+--------+
| SMITH   |    800 |
| AARAV   |   3350 |
| THOMAS  |   1750 |
| ADAMS   |   1100 |
| JAMES   |    950 |
| MILLER  |   1300 |
| NATASHA |   1750 |
+---------+--------+
7 rows in set (0.00 sec)
```

23. *Find the day of the week on which SMITH joined*

mysql> select ename as Name, hiredate, dayName(hiredate) as DayOfWeek from emp where ename = 'SMITH';

```
mysql>
mysql> select ename as Name, hiredate, dayName(hiredate) as DayOfWeek from emp where ename = 'SMITH';
+-------+------------+-----------+
| Name  | hiredate   | DayOfWeek |
+-------+------------+-----------+
| SMITH | 1980-12-17 | Wednesday |
+-------+------------+-----------+
1 row in set (0.00 sec)

mysql>
mysql>
mysql>
```

24. *Find the time of time of the day in which ADAMS joined*

mysql > select ename as Name, hiredate, time(hiredate) as TimeOfDay from emp where ename = 'ADAMS';

```
+-------+------------+-----------+
| Name  | hiredate   | TimeOfDay |
+-------+------------+-----------+
| ADAMS | 1983-01-12 | 00:00:00  |
+-------+------------+-----------+
1 row in set (0.04 sec)
```

25. *Find day of month on which KING joined*

mysql> select ename as Name, hiredate, day(hiredate) as DayOfMonth from emp where ename = 'KING';

```
+------+------------+------------+
| Name | hiredate   | DayOfMonth |
+------+------------+------------+
| KING | 1981-11-17 |         17 |
+------+------------+------------+
1 row in set (0.05 sec)
```

26. *Find out month on which MARTIN joined*

mysql> select ename as Name, hiredate, monthname(hiredate) as MonthJoined from emp where ename = 'MARTIN';

```
+--------+------------+-------------+
| Name   | hiredate   | MonthJoined |
+--------+------------+-------------+
| MARTIN | 1981-09-28 | September   |
+--------+------------+-------------+
1 row in set (0.05 sec)
```

27. *Find out which quarter of the year the employees joined. Display their number and names as well*

```
select empno as EmployeeNumber,
    ename as Name,
    quarter(hiredate) as QuarterJoined
from emp;
```

```
mysql> select empno as EmployeeNumber,
    ->        ename as Name,
    ->        quarter(hiredate) as QuarterJoined
    -> from emp;
+----------------+----------+---------------+
| EmployeeNumber | Name     | QuarterJoined |
+----------------+----------+---------------+
|           6001 | GITA     |             3 |
|           6129 | VRUSHALI |             3 |
|           6473 | SHARMIN  |             3 |
|           6781 | BANDISH  |             3 |
|           7369 | SMITH    |             4 |
|           7415 | AARAV    |             4 |
|           7421 | THOMAS   |             3 |
|           7499 | ALLEN    |             1 |
|           7521 | WARD     |             1 |
|           7566 | JONES    |             2 |
|           7654 | MARTIN   |             3 |
|           7698 | BLAKE    |             2 |
|           7782 | CLARK    |             2 |
|           7788 | SCOTT    |             4 |
|           7839 | KING     |             4 |
|           7840 | FRED     |             3 |
|           7844 | TURNER   |             3 |
|           7876 | ADAMS    |             1 |
|           7888 | Ruhan    |             3 |
|           7889 | Neel     |             3 |
|           7900 | JAMES    |             4 |
|           7902 | FORD     |             4 |
|           7919 | HOFFMAN  |             1 |
|           7920 | GRASS    |             1 |
|           7934 | MILLER   |             1 |
|           7945 | AARUSH   |             1 |
|           7949 | ALEX     |             1 |
|           8433 | NATASHA  |             3 |
|           9400 | SANGITA  |             3 |
|           9473 | SUPRIYA  |             3 |
+----------------+----------+---------------+
30 rows in set (0.06 sec)
```

28. *Retrieve ANALYST records with the hiredate formatted as – 'The 3rd of December 1984'*

select ename as Name,

date_format(hiredate, 'The %D of %M %Y') as FormattedHireDate

from emp

where job = 'ANALYST';

```
mysql>
mysql> select ename as Name,
    ->        date_format(hiredate, 'The %D of %M %Y') as FormattedHireDate
    -> from emp
    -> where job = 'ANALYST';
+---------+--------------------------+
| Name    | FormattedHireDate        |
+---------+--------------------------+
| SCOTT   | The 9th of December 1982 |
| FORD    | The 3rd of December 1981 |
| SUPRIYA | The 19th of July 1981    |
+---------+--------------------------+
3 rows in set (0.00 sec)

mysql>
```

29. *List all names, jobs, and a job classification number, which is to be assigned by you. Translate the value started in each job field to a job classification number. This is to be done as follows-*

A. CLERK

B. MANAGER

C. PRESIDENT

D. OTHER

```
select ename as Name,
    job as Job,
    case
     when job = 'CLERK' then 'A'
     when job = 'MANAGER' then 'B'
     when job = 'PRESIDENT' then 'C'
     else 'D'
    end as JobClassification from emp;
```

```
    -> from emp;
+----------+--------------------+-------------------+
| Name     | Job                | JobClassification |
+----------+--------------------+-------------------+
| GITA     | Compliance officers | D                |
| VRUSHALI | Compliance officers | D                |
| SHARMIN  | Public Relation    | D                |
| BANDISH  | Public Relation    | D                |
| SMITH    | CLERK              | A                |
| AARAV    | CLERK              | A                |
| THOMAS   | CLERK              | A                |
| ALLEN    | SALESMAN           | D                |
| WARD     | SALESMAN           | D                |
| JONES    | MANAGER            | B                |
| MARTIN   | SALESMAN           | D                |
| BLAKE    | MANAGER            | B                |
| CLARK    | MANAGER            | B                |
| SCOTT    | ANALYST            | D                |
| KING     | PRESIDENT          | C                |
| FRED     | vice PRESIDENT     | D                |
| TURNER   | SALESMAN           | D                |
| ADAMS    | CLERK              | A                |
| Ruhan    | Network            | D                |
| Neel     | vice PRESIDENT     | D                |
| JAMES    | CLERK              | A                |
| FORD     | ANALYST            | D                |
| HOFFMAN  | MANAGER            | B                |
| GRASS    | SALESMAN           | D                |
| MILLER   | CLERK              | A                |
| AARUSH   | SALESMAN           | D                |
| ALEX     | MANAGER            | B                |
| NATASHA  | CLERK              | A                |
| SANGITA  | Compliance officers | D                |
| SUPRIYA  | ANALYST            | D                |
+----------+--------------------+-------------------+
30 rows in set (0.00 sec)
```

30. *Display the length of the longest employees name.*

```
mysql> select max(length(ename)) as LongestNameLength from emp;
+----------------------------+
| LongestNameLength |
+---------------------------+
|          8              |
+----------------------+
1 row in set (0.00 sec)
```

31. Write a query to list the length of service of the employees (of the form n years and m months).

```
select ename as Name,
    concat(
      timestampdiff(year, hiredate, curdate()), ' years and ',
      timestampdiff(month, hiredate, curdate())
        - timestampdiff(year, hiredate, curdate()) * 12,
      ' months'
    ) as LengthOfService
from emp;
```

```
+----------+------------------------+
| Name     | LengthOfService        |
+----------+------------------------+
| GITA     | 44 years and 9 months  |
| VRUSHALI | 44 years and 9 months  |
| SHARMIN  | 44 years and 9 months  |
| BANDISH  | 44 years and 9 months  |
| SMITH    | 45 years and 4 months  |
| AARAV    | 44 years and 3 months  |
| THOMAS   | 44 years and 9 months  |
| ALLEN    | 45 years and 2 months  |
| WARD     | 45 years and 1 months  |
| JONES    | 45 years and 0 months  |
| MARTIN   | 44 years and 6 months  |
| BLAKE    | 44 years and 11 months |
| CLARK    | 44 years and 10 months |
| SCOTT    | 43 years and 4 months  |
| KING     | 44 years and 5 months  |
| FRED     | 42 years and 8 months  |
| TURNER   | 44 years and 7 months  |
| ADAMS    | 43 years and 3 months  |
| Ruhan    | 42 years and 8 months  |
| Neel     | 42 years and 8 months  |
| JAMES    | 44 years and 4 months  |
| FORD     | 44 years and 4 months  |
| HOFFMAN  | 44 years and 0 months  |
| GRASS    | 46 years and 2 months  |
| MILLER   | 44 years and 2 months  |
| AARUSH   | 46 years and 2 months  |
| ALEX     | 44 years and 2 months  |
| NATASHA  | 44 years and 9 months  |
| SANGITA  | 44 years and 9 months  |
| SUPRIYA  | 44 years and 9 months  |
+----------+------------------------+
```

32. How many employees who are joined in 1985.

```
mysql> select count(*) as EmployeesJoined1985 from emp where year(hiredate) = 1985;
+----------------------------+
| EmployeesJoined1985 |
+----------------------------+
|            0               |
+----------------------------+
1 row in set (0.03 sec)
```

33. How many employees joined each month in 1985.

```
mysql> select monthname(hiredate) as Month,
    ->      count(*) as EmployeesJoined
    -> from emp
    -> where year(hiredate) = 1985
    -> group by year(hiredate), month(hiredate), monthname(hiredate)
    -> order by month(hiredate);
Empty set (0.03 sec)
```

34. How many employees who are joined in March 1985

```
mysql> select count(*) as EmployeesJoinedMarch1985
    -> from emp
    -> where year(hiredate) = 1985
    ->   and month(hiredate) = 3;
+-------------------------------------+
| EmployeesJoinedMarch1985 |
+-------------------------------------+
|                    0                |
+-------------------------------------+
1 row in set (0.00 sec)
```

35. Find the total sales amount

```
mysql> select sum(sal) as TotalSalesAmount from emp where job = 'SALESMAN';
+------------------------+
| TotalSalesAmount |
+------------------------+
|          9525          |
+------------------------+
1 row in set (0.00 sec)
```

36. Find the customer-wise lowest and highest sales amount

```
select custname,
    min(amount) as LowestSale,
    max(amount) as HighestSale
from sales
group by custname;
```

```
mysql> select custname,
    ->        min(amount) as LowestSale,
    ->        max(amount) as HighestSale
    -> from sales
    -> group by custname;
+----------------+------------+-------------+
| custname       | LowestSale | HighestSale |
+----------------+------------+-------------+
| JOCKSPORTS     |      50.00 |     2400.00 |
| TKB SPORT SHOP |       8.40 |       58.00 |
| VOLLYRITE      |    2300.50 |    16569.00 |
| JUST TENNIS    |      24.00 |      450.00 |
| EVERY MOUNTAIN |      24.00 |     3000.00 |
| K + T SPORTS   |     340.00 |    29000.00 |
| SHAPE UP       |       2.40 |      440.00 |
| WOMENS SPORTS  |     180.00 |      280.00 |
| NORTH WOODS    |     440.00 |     4800.00 |
+----------------+------------+-------------+
9 rows in set (0.04 sec)
```

37. Find product-wise lowest, highest and total sales.

select prodname as Product,
min(amount) as LowestSale,
max(amount) as HighestSale,
sum(amount) as TotalSales
from sales
group by prodname
order by prodname;

```
mysql> select prodname as Product,
    ->        min(amount) as LowestSale,
    ->        max(amount) as HighestSale,
    ->        sum(amount) as TotalSales
    -> from sales
    -> group by prodname
    -> order by prodname;
+-------------------------+------------+-------------+------------+
| Product                 | LowestSale | HighestSale | TotalSales |
+-------------------------+------------+-------------+------------+
| ACE TENNIS BALLS-3 PACK |       8.40 |     3306.00 |    5426.90 |
| ACE TENNIS BALLS-6 PACK |     250.00 |     5600.00 |   10056.80 |
| ACE TENNIS NET          |      50.00 |    29000.00 |   29398.00 |
| ACE TENNIS RACKET I     |      35.00 |    16569.00 |   22144.00 |
| ACE TENNIS RACKET II    |      84.00 |     4500.00 |    8809.50 |
| RH: "GUIDE TO TENNIS"   |      34.00 |     1703.40 |    2417.40 |
| SB ENERGY BAR-6 PACK    |       2.40 |     2400.00 |    3586.40 |
| SB VITA SNACK-6 PACK    |     400.00 |     1200.00 |    2200.00 |
| SP JUNIOR RACKET        |    1500.00 |     2500.00 |    4000.00 |
| SP TENNIS RACKET        |      24.00 |     4800.00 |    7224.00 |
+-------------------------+------------+-------------+------------+
10 rows in set (0.04 sec)
```

38. Find department-wise average salary for all the departments employing more than three employees

mysql> select deptno as Department, avg(sal) as AverageSalary from emp group by deptno having count(*) > 3;

```
+------------+---------------+
| Department | AverageSalary |
+------------+---------------+
|       10 |    2770.0000 |
|       20 |    2421.8750 |
|       30 |    1852.0833 |
+------------+---------------+
3 rows in set (0.00 sec)
```

39. Find the customer-wise total sales for all the customers except 'TKB SPORT SHOP' who came to purchase various sports items maximum four times.

```
mysql> select custname, sum(amount) as TotalSales
   -> from sales
   -> where custname <> 'TKB SPORT SHOP'
   -> group by custname
   -> having count(*) <= 4;
+---------------+------------------------+
| custname      | TotalSales             |
+---------------+------------------------+
| VOLLYRITE     |   27775.50             |
| SHAPE UP      |    700.40              |
| WOMENS SPORTS |    710.00 |
| NORTH WOODS   |   6400.00 |
+---------------+---------------------+
4 rows in set (0.00 sec)
```

40. Display the highest, lowest, sum and average salary for all employees. Label the columns appropriately.

```
select max(sal) as HighestSalary,
    min(sal) as LowestSalary,
    sum(sal) as TotalSalary,
    avg(sal) as AverageSalary
from emp;
+---------------+--------------+-------------+-------------------------------+
| HighestSalary | LowestSalary | TotalSalary | AverageSalary |
+---------------+--------------+-------------+-------------------------------+
|        5000 |        800 |      63700 |    2123.3333                    |
+---------------+--------------+-------------+-------------------------------+
1 row in set (0.00 sec)
```

41. Modify the above query and display the output for each job type.

```
select job as JobType,
    max(sal) as HighestSalary,
    min(sal) as LowestSalary,
```

sum(sal) as TotalSalary,

avg(sal) as AverageSalary

from emp
group by job;

```
mysql> select job as JobType,
    ->        max(sal) as HighestSalary,
    ->        min(sal) as LowestSalary,
    ->        sum(sal) as TotalSalary,
    ->        avg(sal) as AverageSalary
    -> from emp
    -> group by job;
+---------------------+---------------+--------------+-------------+---------------+
| JobType             | HighestSalary | LowestSalary | TotalSalary | AverageSalary |
+---------------------+---------------+--------------+-------------+---------------+
| Compliance officers |          1750 |         1750 |        5250 |     1750.0000 |
| Public Relation     |          1750 |         1750 |        3500 |     1750.0000 |
| CLERK               |          3350 |          800 |       11000 |     1571.4286 |
| SALESMAN            |          2575 |         1250 |        9525 |     1587.5000 |
| MANAGER             |          4150 |         1250 |       13675 |     2735.0000 |
| ANALYST             |          3000 |         1750 |        7750 |     2583.3333 |
| PRESIDENT           |          5000 |         5000 |        5000 |     5000.0000 |
| vice PRESIDENT      |          5000 |         1000 |        6000 |     3000.0000 |
| Network             |          2000 |         2000 |        2000 |     2000.0000 |
+---------------------+---------------+--------------+-------------+---------------+
9 rows in set (0.00 sec)
```

42. List names of people who have salary less than the average salary for dept 20

mysql> select ename from emp where deptno = 20 and sal < (select avg(sal) from emp where deptno = 20);

```
+---------+
| ename   |
+---------+
| SHARMIN |
| BANDISH |
| SMITH   |
| ADAMS   |
+---------+
4 rows in set (0.06 sec)
```

43. Find the average annual salary per job in each department.

select deptno as Department,

job as JobType,

*avg(sal*12) as AvgAnnualSalary*

from emp
group by deptno, job;

```
mysql> select deptno as Department,
    ->        job as JobType,
    ->        avg(sal*12) as AvgAnnualSalary
    -> from emp
    -> group by deptno, job;
+------------+---------------------+-----------------+
| Department | JobType             | AvgAnnualSalary |
+------------+---------------------+-----------------+
|         30 | Compliance officers |      21000.0000 |
|         50 | Compliance officers |      21000.0000 |
|         20 | Public Relation     |      21000.0000 |
|         20 | CLERK               |      11400.0000 |
|         10 | CLERK               |      25600.0000 |
|         30 | SALESMAN            |      19050.0000 |
|         20 | MANAGER             |      35700.0000 |
|         30 | MANAGER             |      33000.0000 |
|         10 | MANAGER             |      29400.0000 |
|         20 | ANALYST             |      36000.0000 |
|         10 | PRESIDENT           |      60000.0000 |
|         20 | vice PRESIDENT      |      60000.0000 |
|         60 | Network             |      24000.0000 |
|         60 | vice PRESIDENT      |      12000.0000 |
|         30 | CLERK               |      16200.0000 |
|         50 | ANALYST             |      21000.0000 |
+------------+---------------------+-----------------+
16 rows in set (0.00 sec)
```

44. *Count the number of people in department 30 who receive a salary and the number of people who receive a commission*

```
select deptno as Department,
    count(sal) as SalaryCount,
    count(comm) as CommissionCount
from emp
where deptno = 30
group by deptno;
```

```
+------------+-------------+-----------------------------+
| Department | SalaryCount | CommissionCount |
+------------+-------------+-----------------------------+
|       30 |        12 |           7                       |
+------------+-------------+-----------------------------+
1 row in set (0.00 sec)
```

45. *Compute the average, minimum and maximum salaries of these groups of employees having job as Clerk or manager, Display the job as well*

```
select job as JobType,
    avg(sal) as AverageSalary,
    min(sal) as MinimumSalary,
    max(sal) as MaximumSalary
from emp
where job in ('CLERK', 'MANAGER')
group by job;
```

```
+---------+---------------+---------------+---------------+
| JobType | AverageSalary | MinimumSalary | MaximumSalary |
+---------+---------------+---------------+---------------+
| CLERK   |    1571.4286 |         800 |        3350 |
| MANAGER |    2735.0000 |        1250 |        4150 |
+---------+---------------+---------------+---------------+
2 rows in set (0.00 sec)
```

46. Write an SQL command that displays 2nd highest salary paid

```
select ename, sal
from emp
where sal = (
  select max(sal)
  from emp
  where sal < (select max(sal) from emp)
);
```

```
+---------------------+
| SecondHighestSalary |
+---------------------+
|              4150 |
+---------------------+
1 row in set (0.00 sec)
```

47. Write a query to find the employees who are earning the maximum salary in their departments.

```
select ename, deptno, sal
from emp e
where sal = (
  select max(sal)
  from emp
  where deptno = e.deptno
);
```

```
+----------+--------+------+
| ename    | deptno | sal  |
+----------+--------+------+
| VRUSHALI |     50 | 1750 |
| KING     |     10 | 5000 |
| FRED     |     20 | 5000 |
| Ruhan    |     60 | 2000 |
| HOFFMAN  |     30 | 4150 |
| SANGITA  |     50 | 1750 |
| SUPRIYA  |     50 | 1750 |
+----------+--------+------+
7 rows in set (0.00 sec)
```

48. Write a query to find the salesman number (repid) who has achieved the maximum total sales among the entire salesman.

```
mysql> select repid, sum(amount) as TotalSales
   -> from sales
   -> group by repid
   -> order by TotalSales desc
   -> limit 1;
+-------+------------+
| repid | TotalSales |
+-------+------------+
|  7844 |   58050.90 |
+-------+------------+
1 row in set (0.01 sec)
```

49. . List the highest salary paid for each job.

```
mysql> select job as JobType,
   ->      max(sal) as HighestSalary
   -> from emp
   -> group by job;
+---------------------+---------------+
| JobType             | HighestSalary |
+---------------------+---------------+
| Compliance officers |          1750 |
| Public Relation     |          1750 |
| CLERK               |          3350 |
| SALESMAN            |          2575 |
| MANAGER             |          4150 |
| ANALYST             |          3000 |
| PRESIDENT           |          5000 |
| vice PRESIDENT      |          5000 |
| Network             |          2000 |
+---------------------+---------------+
9 rows in set (0.00 sec)
```

50. Find the most recently hired employee in each department.

```
mysql> select ename, deptno, hiredate
   -> from emp e
   -> where hiredate = (
   ->    select max(hiredate)
   ->    from emp
   ->    where deptno = e.deptno
   -> );
```

```
+----------+--------+------------+
| ename    | deptno | hiredate   |
+----------+--------+------------+
| VRUSHALI |     50 | 1981-07-19 |
| FRED     |     20 | 1983-08-09 |
| Ruhan    |     60 | 1983-08-09 |
| Neel     |     60 | 1983-08-09 |
| HOFFMAN  |     30 | 1982-03-24 |
| MILLER   |     10 | 1982-01-23 |
| SANGITA  |     50 | 1981-07-19 |
| SUPRIYA  |     50 | 1981-07-19 |
+----------+--------+------------+
8 rows in set (0.00 sec)
```

51. *In which year did most people join the company? Display the year and the number of employees.*

```
mysql> select year(hiredate) as YearJoined,
   ->       count(*) as NumEmployees
   -> from emp
   -> group by year(hiredate)
   -> order by NumEmployees desc
   -> limit 1;
+------------+--------------+
| YearJoined | NumEmployees |
+------------+--------------+
|       1981 |           19 |
+------------+--------------+
1 row in set (0.00 sec)
```

52. *Write a query to display employee name whose name occurs only once in the table.*

```
mysql> select ename from emp group by ename having count(*) = 1;

+----------+
| ename    |
+----------+
| GITA     |
| VRUSHALI |
| SHARMIN  |
| BANDISH  |
| SMITH    |
| AARAV    |
| TURNER   |
| ADAMS    |
| Ruhan    |
| Neel     |
| JAMES    |
| SUPRIYA  |
+----------+
30 rows in set (0.00 sec)
```

53. . Write a query to display all the details from dept table along with the no. of employee working in each dept.

```
select d.deptno,
    d.dname,
    d.loc,
    count(e.empno) as NumEmployees
from dept d
left join emp e
    on d.deptno = e.deptno
group by d.deptno, d.dname, d.loc;
+--------+------------+----------+--------------+
| deptno | dname      | loc      | NumEmployees |
+--------+------------+----------+--------------+
|     10 | ACCOUNTING | NEW YORK |            5 |
|     20 | RESEARCH   | DALLAS   |            8 |
|     30 | SALES      | CHICAGO  |           12 |
|     40 | OPERATIONS | BOSTON   |            0 |
|     50 | HR         | PUNE     |            3 |
|     60 | IT         | BARODA   |            2 |
+--------+------------+----------+--------------+
6 rows in set (0.05 sec)
```

54. . Find out which department does not have any employees.

```
mysql> select d.deptno,
   ->      d.dname,
   ->      d.loc
   -> from dept d
   -> left join emp e
   ->      on d.deptno = e.deptno
   -> where e.empno is null;
+--------+------------+--------+
| deptno | dname      | loc    |
+--------+------------+--------+
|     40 | OPERATIONS | BOSTON |
+--------+------------+--------+
1 row in set (0.00 sec)
```

Q. 55. . List out the no. of employees joined in every month in ascending order.

```
mysql> select month(hiredate) as MonthJoined,
   ->      count(*) as NumEmployees
```

```
   -> from emp
   -> group by month(hiredate)
   -> order by MonthJoined asc;
+-------------+--------------+
| MonthJoined | NumEmployees |
+-------------+--------------+
|           1 |            3 |
|           2 |            4 |
|           3 |            1 |
|           4 |            1 |
|           5 |            1 |
|           6 |            1 |
|           7 |            8 |
|           8 |            3 |
|           9 |            2 |
|          11 |            1 |
|          12 |            5 |
+-------------+--------------+
11 rows in set (0.04 sec)
```